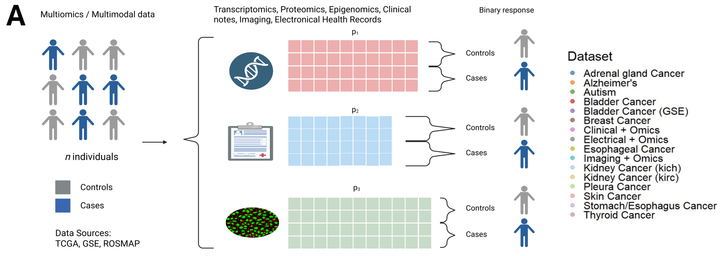
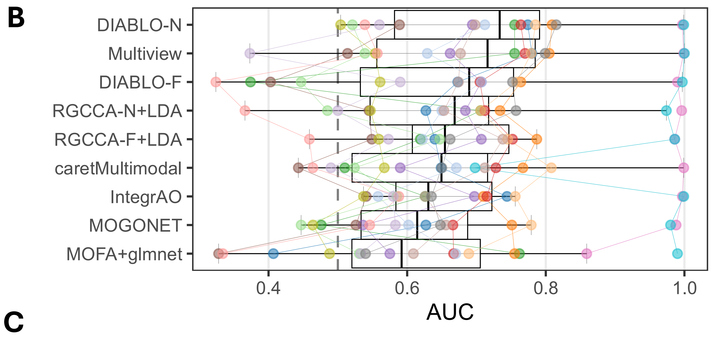
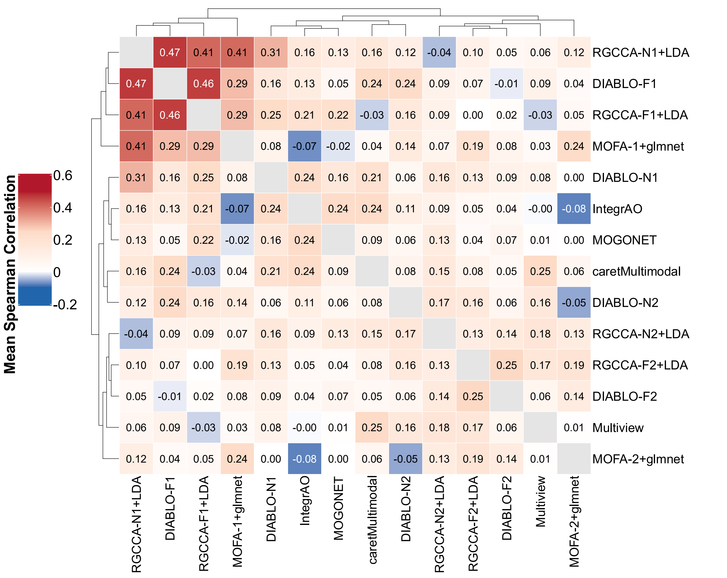
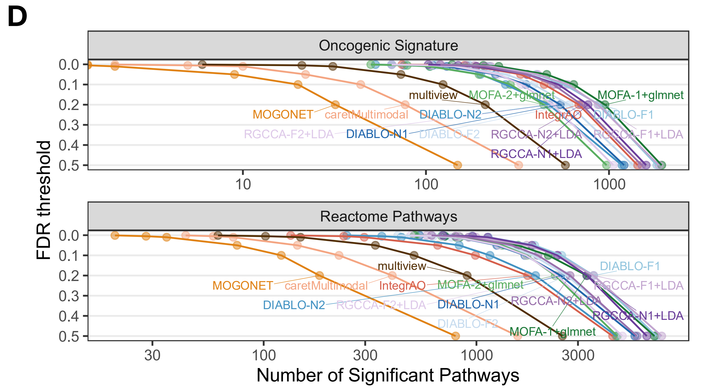
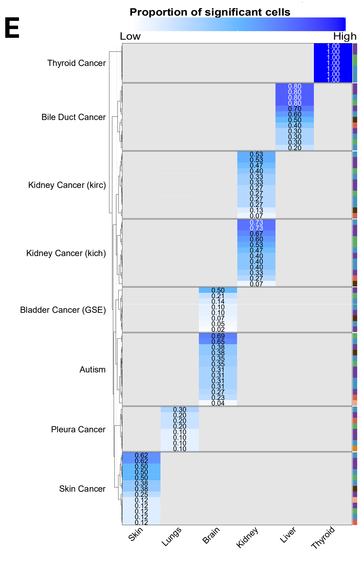
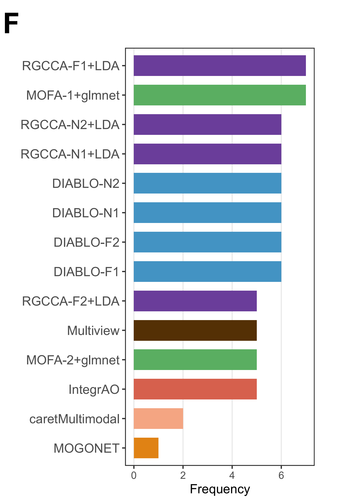

A
Multiomics / Multimodal data
n individuals
Controls
Cases
Data Sources:
TCGA, GSE, ROSMAP
Transcriptomics, Proteomics, Epigenomics, Clinical notes, Imaging, Electronical Health Records
Binary response
Controls
Cases
Dataset
Adrenal gland Cancer
Alzheimer's
Autism
Bladder Cancer
Bladder Cancer (GSE)
Breast Cancer
Clinical + Omics
Electrical + Omics
Esophageal Cancer
Imaging + Omics
Kidney Cancer (kich)
Kidney Cancer (kirc)
Pleura Cancer
Skin Cancer
Stomach/Esophagus Cancer
Thyroid Cancer

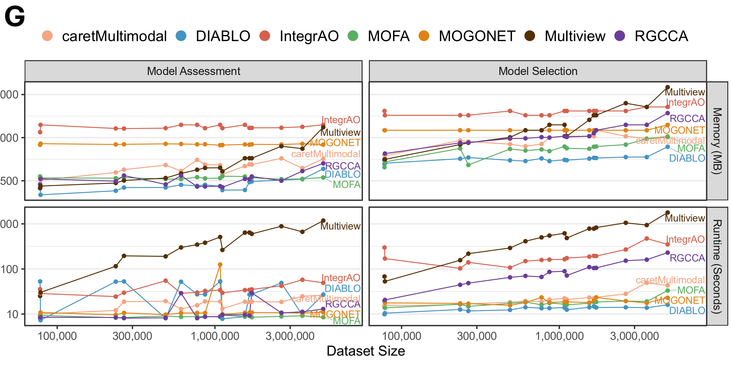

B
DIABLO-N
Multiview
DIABLO-F
RGCCA-N+LDA
RGCCA-F+LDA
caretMultimodal
IntegrAO
MOGONET
MOFA+glmnet
0.4
0.6
0.8
1.0
AUC


C
Mean Spearman Correlation
0.6
0.4
0.2
0
-0.2
RGCCA-N1+LDA
DIABLO-F1
RGCCA-F1+LDA
MOFA-1+glmnet
DIABLO-N1
IntegrAO
MOGONET
caretMultimodal
DIABLO-N2
RGCCA-N2+LDA
RGCCA-F2+LDA
DIABLO-F2
Multiview
MOFA-2+glmnet


D
Oncogenic Signature
Reactome Pathways
FDR threshold
0.0
0.1
0.2
0.3
0.4
0.5
10
100
1000
30
300
3000
Number of Significant Pathways
multiview
MOFA-2+glmnet
MOFA-1+glmnet
MOGONET
caretMultimodal
DIABLO-N2
IntegrAO
DIABLO-F1
RGCCA-F2+LDA
DIABLO-N1
DIABLO-F2
RGCCA-N2+LDA
RGCCA-F1+LDA
RGCCA-N1+LDA


E
Proportion of significant cells
Low
High
Thyroid Cancer
Bile Duct Cancer
Kidney Cancer (kirc)
Kidney Cancer (kich)
Bladder Cancer (GSE)
Autism
Pleura Cancer
Skin Cancer
Skin
Lungs
Brain
Kidney
Liver
Thyroid


F
RGCCA-F1+LDA
MOFA-1+glmnet
RGCCA-N2+LDA
RGCCA-N1+LDA
DIABLO-N2
DIABLO-N1
DIABLO-F2
DIABLO-F1
RGCCA-F2+LDA
Multiview
MOFA-2+glmnet
IntegrAO
caretMultimodal
MOGONET
0
2
4
6
Frequency


G
caretMultimodal
DIABLO
IntegrAO
MOFA
MOGONET
Multiview
RGCCA
Model Assessment
Model Selection
Memory (MB)
Runtime (Seconds)
2000
1000
500
100
10
100,000
300,000
1,000,000
3,000,000
Dataset Size